Takumi Endo

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https://takuendo.github.io/

PROFILE

- CFD engineer and Ph.D. candidate at TUM, developing deep-learning methods to make LES affordable for vehicle aerodynamics.
- Wall modeling: GAN-based wall model coupled online with OpenFOAM, predicting separated flows at about 2% of the cost of wall-resolved LES.
- Super-resolution: geometry-aware CycleGAN that corrects coarse LES of the Ahmed body wake, reducing the recirculation-length error from 12% to 2%.
- 4 years of industrial CFD experience in motorsport external aerodynamics (TotalSim Japan), from meshing and case setup to post-processing.

EDUCATION

Jan. 2026 – present:	Ph.D. candidate in Chair of Aerodynamics and Fluid Mechanics TUM School of Engineering and Design – Technical University of Munich at Munich, Germany - Development of a Wall Model for Large Eddy Simulation with Generative Adversarial Networks for Applications in Vehicle Aerodynamics - Fully funded by Studienstiftung des deutschen Volkes
Apr. 2025 – Dec. 2025:	Ph.D. candidate in Mechanical Engineering – The University of Tokyo at Tokyo, Japan - Funded by JST SPRING; suspended upon transfer to TUM
Apr. 2023 – Mar. 2025:	Master of Engineering in Mechanical Engineering – The University of Tokyo at Tokyo, Japan (GPA: 3.84/4) - Developed a GAN-based wall model for LES, integrating machine learning into OpenFOAM CFD workflows - Built surrogate models for wind-assisted ship propulsion systems (rigid sails, rotor sails) using CFD data
Sep. 2023 – Feb. 2024:	Exchange program in Aeronautical Engineering – Politecnico di Milano at Milan, Italy - Conducted CFD analysis of a Formula 1 DRS system to optimize aerodynamic performance
Apr. 2019 – Mar. 2023:	Bachelor of Engineering in Mechanical Engineering – The University of Tokyo at Tokyo, Japan

INTERNSHIP

Oct. 2022 – Feb. 2026:	CFD Engineer at TotalSim Japan Inc. at Tokyo, Japan - Conducted external aerodynamic analyses of motorsport cars and racing motorcycles - Set up simulation cases and meshes in OpenFOAM - Post-processed results using ParaView, Excel, and Python/Linux scripts

EXTRACURRICULAR ACTIVITIES

Nov. 2023 - Mar. 2024: Aerodynamics Engineer at Polimi Motorcycle Factory (PMF)
- Belonged to Aerodynamics department in the official team of Politecnico di Milano competing in MotoStudent
- Conducted steady simulations, analyzed results, and developed rotating models of transient simulations

JOURNAL PUBLICATIONS

Aug. 2026: Endo, T. and Indinger, T., "A Patch-Based Adversarial Wall Model for Large-Eddy Simulation of Separated Flows", Physics of Fluids, under review.

Dec. 2025: Endo, T. and Guzelbulut, C., “Performance Analysis of Interacting Rigid Wind Sails and Rotor Sails by Kriging Surrogate Model", Journal of ETA Maritime Science, 2025.

CONFERENCE PRESENTATIONS (peer reviewed)

Jan. 2027: Endo, T. and Indinger, T., "GAN-Based Wall Model for Large Eddy Simulation of Periodic Hill Flow", AIAA SciTech Forum 2027, Orlando, USA, accepted for oral presentation (abstract-reviewed).

Sep. 2025: Endo, T., et al., "Development of GANs-based Wall Model for Large Eddy Simulation of Wall-Bounded Flow", 15th International ERCOFTAC Symposium on Engineering Turbulence Modelling and Measurements (ETMM-15), 2025, Dubrovnik, Croatia, Oral presentation.

AWARDS

Mar. 2026: Doctoral Scholarship by Studienstiftung des deutschen Volkes

Apr. 2025: Doctoral Scholarship by JST-SPRING

Mar. 2025: Outstanding Presentation Award at the 40th IIS TSFD Symposium, Tokyo, Japan

Aug. 2023: Continental UTokyo-IIS Global Engineering Fellowship – The University of Tokyo and Continental Japan

SKILLS

- CFD: LES / WMLES, wall modeling, external aerodynamics (vehicles, motorsport)
- OpenFOAM: 4 years — solver customization, wall-model integration, case setup, automation
- ML for CFD: GAN-based wall models coupled online with OpenFOAM; unpaired super-resolution of LES with geometry-aware CycleGAN; PyTorch
- Programming: Python (4 years), C++ (OpenFOAM), MATLAB/Simulink (2 years), Bash/Linux
- Tools / HPC: ParaView, Git, LRZ supercomputing resources
- Languages: Japanese (native), English (fluent), German (beginner)